

Project Plan: Mixture of Experts Framework for Personalised Characterization of the Insulin Synthesis System

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1 Introduction

In the medical field, there is an increasing focus on individualised treatment of patients. A deeper understanding of a patient’s metabolic system state enables better clinical assessment, which in turn allows for personalised therapeutic interventions. This extends to real-time systems such as insulin pumps, where accurate patient-specific models can enhance treatment speed and efficacy.

A central challenge in this domain is the heterogeneity across patients: creating a single model that adequately captures the metabolic dynamics of diverse individuals remains an open problem. The Mixture of Experts (MoE) paradigm offers a principled approach to this challenge by allowing specialised sub-models (experts) to handle distinct patient subgroups or operating regimes, coordinated by a gating mechanism that routes inputs to the most appropriate expert.

This project investigates how a Mixture of Experts framework can be applied to enable personalised characterisation of the insulin synthesis system in the human pancreas, with the ultimate goal of supporting patient-specific treatment strategies.

2 Problem Statement

The human metabolic system, and in particular the insulin synthesis pathway, can be mathematically modelled as a dynamical system. However, the available clinical data is typically sparse: measurements are infrequent, noisy, and vary significantly across patients. This sparsity makes traditional single-model identification approaches unreliable, especially when inter-patient variability in physiological parameters is high.

The core research question driving this project is:

How can a Mixture of Experts model approach allow for personalised characterisation of the Insulin Synthesis System in the pancreas, given sparse and heterogeneous patient data?

To address this, the project will investigate the Mixture of Experts framework as a means to discover inter-variable parameters and conditions that distinguish patient subgroups, enabling robust identification of individual patient dynamics from limited data samples.

3 Requirements

The following requirements guide the design and implementation of the system:

- R1. Mixture of Experts architecture** — The framework must implement a modular MoE structure in which individual experts can be independently designed, trained, and evaluated.
- R2. Initial multi-PID setup** — The baseline expert architecture shall employ multiple PID controllers as individual experts, reflecting the control-theoretic nature of metabolic regulation.
- R3. One-time learning process** — The system must support a one-time training phase over a diverse patient population, after which expert selection and personalisation can occur without full retraining.
- R4. Sparse data identification** — The framework must be capable of identifying patient-specific dynamics from a sparse number of data samples, leveraging the shared structure encoded across experts.

4 Project Objectives and Deliverables

4.1 Objectives

- O1.** Design and implement a Mixture of Experts framework with modular, interchangeable expert components.
- O2.** Develop a multi-PID-based expert configuration as the initial baseline approach.
- O3.** Evaluate the framework’s ability to personalise metabolic characterisation under data sparsity.
- O4.** Compare model performance against synthetic data and the raw Tokyo dataset.

4.2 Deliverables

- D1.** A functional Mixture of Experts software framework with documented API and modular expert interface.
- D2.** Performance evaluation report including:
 - (a) Validation against synthetic data with known ground truth.
 - (b) Evaluation on the raw Tokyo clinical dataset.
- D3.** Individual research contributions (see Section 5.2).
- D4.** Final presentation and project documentation.

5 Methodology

5.1 Collective Work

The shared methodology follows a structured engineering process:

- 1. **Preliminary research** — Literature review on Mixture of Experts models, PID-based metabolic control, and patient-specific system identification to establish a common theoretical foundation.
- 2. **System requirements specification** — Formalise the functional and non-functional requirements for the MoE framework, including interfaces, data formats, and performance criteria.
- 3. **Architecture design and implementation** — Design the general MoE architecture with a clear separation between the gating mechanism, expert modules, and evaluation pipeline. Implement the multi-PID expert approach as baseline.
- 4. **Verification and validation** — Verify correctness against synthetic datasets with known parameters. Validate on the Tokyo clinical dataset and assess personalisation quality.
- 5. **Results analysis and recommendations** — Synthesise findings, identify limitations, and formulate recommendations for future work.

5.2 Individual Contributions

In addition to the collective work, each team member pursues an individual research direction that extends or complements the shared framework. These contributions are detailed in Section 6 under each member’s role description.

6 Team Roles and Responsibilities

6.1 Gabriele Salvo — Team Manager / Contact Person

Main focus: Coordination and communication.

Responsibilities:

- Main point of contact with supervisors, clients, and stakeholders.
- Assigns tasks, tracks deadlines, and ensures the project stays on schedule.
- Schedules meetings and maintains team alignment on goals and progress.
- Handles progress updates, status reports, and conflict resolution.

Individual research contribution: Black-box system identification of the human metabolic response. Rather than modelling the internal biochemical pathways explicitly, this approach treats the human body as a black box: given measurable inputs (e.g. glucose intake, insulin dosage) and observable outputs (e.g. blood glucose level over time), the goal is to identify a transfer function or equivalent input-output model that characterises an individual patient's metabolic response. This data-driven approach complements the mechanistic MoE framework by providing an alternative expert type that can be fitted from minimal clinical observations without requiring detailed physiological assumptions.

6.2 Giulio Ginami — Project Lead / Strategist

Main focus: Planning and direction.

Responsibilities:

- Develops the overall project strategy and defines milestones.
- Breaks the project into actionable work packages.
- Makes key technical and strategic decisions with team input.
- Ensures all work aligns with the stated objectives.
- Oversees quality and coherence of the final deliverables.

Individual research contribution: Lead researcher on the Mixture of Experts architecture. Investigates gating mechanism design, expert routing strategies, and the theoretical properties of MoE models as applied to heterogeneous patient populations.

6.3 Rens Scheepers — Technical Specialist / Executor

Main focus: Implementation.

Responsibilities:

- Handles core technical and implementation tasks.
- Solves technical problems related to modelling and simulation.
- Ensures functionality and performance of the software framework.
- Supports other team members with technical aspects of their work.

Individual research contribution: Lead in metabolic modelling. Develops and refines the mathematical models of insulin synthesis dynamics that serve as the basis for individual experts within the MoE framework.

6.4 Sem van der Worp — Research and Documentation Lead

Main focus: Information and records.

Responsibilities:

- Conducts background research and gathers relevant data and literature.
- Prepares reports, documentation, and meeting minutes.
- Organises shared files, references, and version-controlled documents.
- Supports presentations with evidence, analysis, and visual materials.

Individual research contribution: Lead in software documentation and reproducibility. Ensures that the codebase, experimental setup, and evaluation procedures are thoroughly documented and reproducible by third parties. Ensures modular software compatibility among expert approaches in the Mixture of Experts framework. Ensures framework integration with existing modelling architecture. Research into Bayesian interpretation of an insulin synthesis expert. Focussing on the physiology behind the insulin production process of the Islets of Langerhans. Considering a networked system of agents achieving a common goal by the principle of free energy minimization. This approach allows for incorporation of mathematical prior information while allowing for updated beliefs based on new available information described by confidence bounds. The expert allows for dynamic independent updates at every stage of the model.

7 Timeline

The project is structured across two quartiles. The following Gantt chart provides an indicative schedule; exact week numbers will be confirmed with the project supervisor.

7.1 Key Milestones

Milestone	Target Week	Description
M1	Week 4	Requirements and literature review complete
M2	Week 7	MoE architecture design finalised
M3	Week 10	Quartile 1 presentation; baseline implementation ready
M4	Week 14	Individual research contributions integrated
M5	Week 17	Evaluation on Tokyo dataset complete
M6	Week 20	Final report and presentation delivered

Table 1: Key project milestones.

8 Communication and Collaboration

The team uses WhatsApp for day-to-day communication and scheduling, and Teams for shared documents and Github for deliverables. Decisions affecting project direction are made during weekly team meetings and documented in meeting minutes. All members are expected to respond to messages within 24 hours on weekdays.

Team Manager, serves as the single point of contact with the project supervisor and any external stakeholders. Progress updates are communicated proactively, and any anticipated delays are flagged at least 48 hours in advance so work can be redistributed.

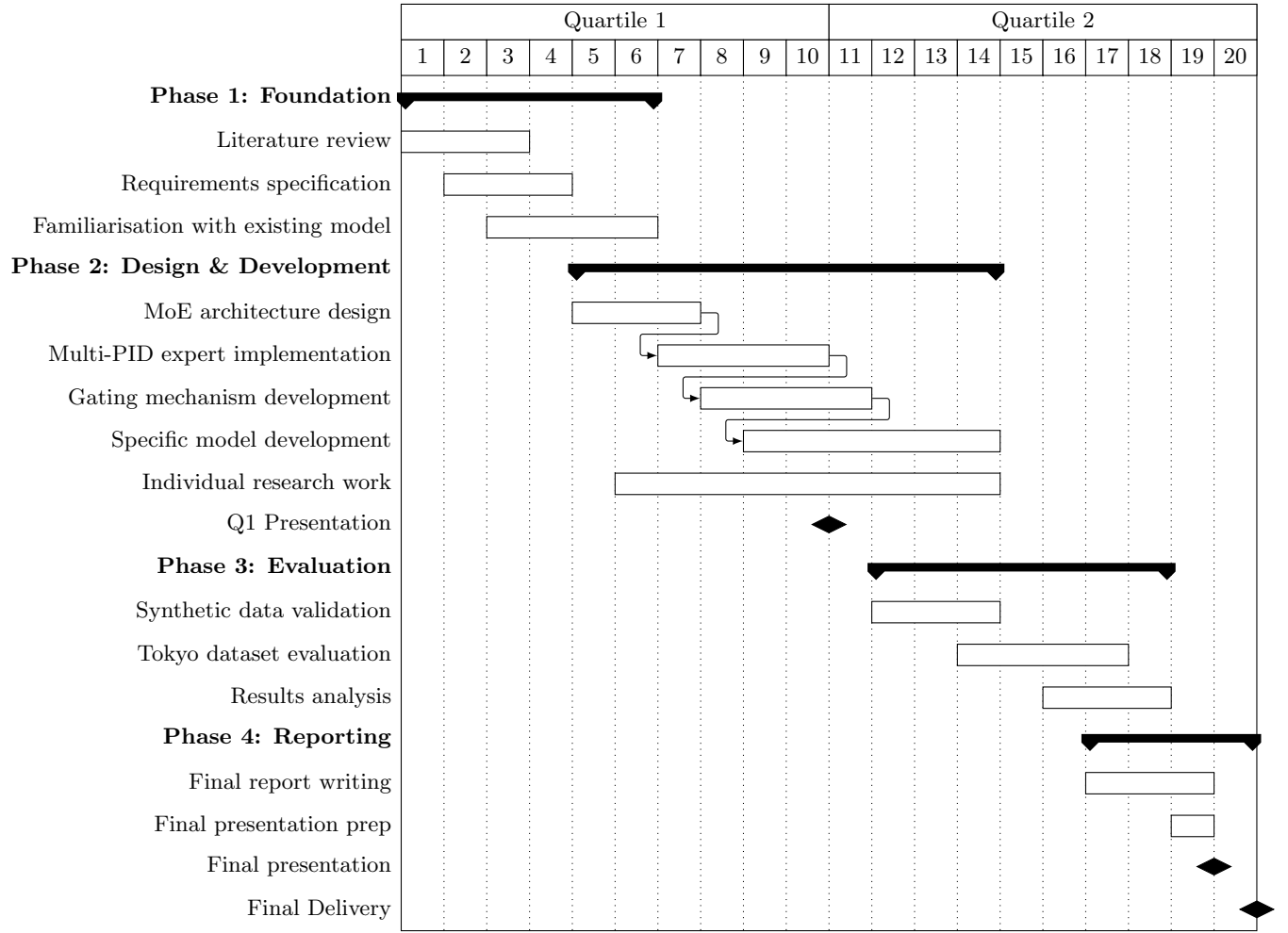


Figure 1: Project timeline (indicative). Week numbers are subject to confirmation with the project supervisor.

9 Risk Management

Risk	Likelihood	Mitigation
Data sparsity limits model identification	High	Augment with synthetic data; design experts that are robust to limited samples.
Integration issues between expert modules	Medium	Define clear interfaces early; continuous integration testing from week 7 onwards.
Scope creep from individual research	Medium	Time-box individual work; weekly check-ins to ensure alignment with project goals.
Team member unavailability	Low	Cross-training on all modules; documented codebase enables handover.
Tokyo dataset quality or access issues	Low	Identify backup datasets during literature review phase.

Table 2: Risk register with mitigation strategies.